Introduction

In the world of soccer analytics, xG (expected goals) is frequently the metric used for evaluating shot quality. Based on previous data, xG represents the probability that a shot taken from that position, in similar contexts, will result in a goal, and models that attempt to calculate xG typically focus on the shot itself and the corresponding characteristics of the shot context; for example, where on the pitch the shot was taken, the angle of the shot, the quality of the goalkeeper, etc. What is not as well understood, and what we are attempting to answer, is if there are certain tactical and lineup conditions leading up to the shot that could be systematically associated with producing high-quality shots.

In short, our goal for this project is to step back and take a possession-level view of shot chance creation. Instead of analyzing all of the details of the specific shot itself, what if we looked at the sequence of possession(s) that lead up to the shot? There are a number of questions we can answer by taking this possession-level approach. Do certain lineup combinations produce more "dangerous" possessions than others on average? Are some possession types higher quality than others?

Data Source

All data was taken from the StatsBomb open data repository, provided through the Trilemma Foundation. It covers over 3,400 matches across different competitions and leagues (primarily European). Some examples include La Liga, Premier League, FIFA World Cup, UEFA Euro tournaments, and the Women's World Cup. Some characteristics of the dataset include:

- Over 12 million individual event records (passes, shots, carries, etc)
- 360-degree spatial tracking data on a smaller subset of matches
- Lineup and formation data, including timing of substitutions
- Player-level statistics, which enabled archetype clustering
- Match metadata (competition, season, home/away teams, etc)

Data Description and EDA

First, a possession can be defined as a continuous, unbroken series of events that one team makes before the ball changes sides. Statsbomb actually tracks possessions by integer sequentially per match so that every event can be linked back to its respective possession. As mentioned before, we were able to capture over 640,000 possessions across the dataset. An example of a possession record would contain match and team identifiers, period and possession numbers, and outcome metrics like total xG generated, number of shots, and possession duration. And as will be further explained, we also added calculated variables to the

possession-level data: possession type, offensive/defensive lineup composition, and archetype clusters.

After we filtered down to the possession level, a few key characteristics of the dataset emerged:

- Shot rate: 12.1% of possessions produced at least one shot.
- XG distribution: Within the shot-producing possessions, the total xG had a mean of 0.116, with a maximum of 3.2. The distribution has a heavy right skew, with a majority of possessions producing a single low-xG shot.
- Possession duration: The median length of possession is roughly 16 seconds. Shot-producing possessions are slightly longer (.38min vs .31min) than non-shot-producing possessions
- Shot rate by possession type: Shot rates vary by possession type. For example, the “Middle Third – Short" type produces shots at a rate of 6.3%, while “Defensive Third – Patient" produces shots at a rate of 23.7%

We also addressed some data quality issues that arose from cleaning and processing:

- In the case of a player substituting mid-possession (these were rare events that did not meaningfully impact the models), we attributed the lineup to the one at the start of the possession
- Teams switch direction at halftime (and per half of extra time if needed). We normalized coordinates and inferred direction from each team’s mean x-position by half, with a threshold of x>=60 on a 120-unit field. This indicted an attack toward the higher end.

Possession Classification

Many definitions of player archetypes exist, but there are few for possessions as a whole. Could we develop a simple taxonomy that groups different types of possessions together? An initial approach aimed to use empirically-derived cluster labels (“Sustained Buildup”, “Fast Counter”, etc), but the results proved to be too noisy and difficult to interpret. The current framework is much simpler, while still giving clean analytic results. Possessions are classified along two dimensions: Start Zone and Progression Style.

Start Zone (where the possession begins) is either:

- Defensive Third: the possession starts in the 40 or less yards on the team’s own half
- Middle Third: the possession starts in the middle 40 yards of the field
- Attacking Third: the possession starts in the last 40 yards of the field, closer to the opposing team’s goal

Progression Style (how the possession develops):

- Direct: Possession gains at least 40 yards on the field
- Patient: Possession lasts 30 or more seconds, and does not meet the Direct threshold
- Short: All remaining possessions.

It is important to note that Progression Style is in priority order, meaning that a possession that gains at least 40 yards and lasts at least 30 seconds is *direct,* not patient. A patient possession should be one that takes a long time to build up without gaining much in the way of pitch progression. It would appear that there should be nine possession types, but since “Attacking Third – Direct" is impossible, these exceedingly rare cases were excluded as they were likely data artifacts, and thus there are eight possession types for analysis: “Defensive Third – Direct", “Defensive Third – Patient", “Defensive Third – Short", “Middle Third – Direct", “Middle Third – Patient", “Middle Third – Short", “Attacking Third – Patient", and “Attacking Third – Short".

These possession types did produce meaningful differences when it came to shot creation and xG, and appear to be tactically meaningful as well:

- As mentioned before, shot rates range from 6.3% (Middle Third - Short) to 23.7% (Defensive Third - Patient). The high shot rate for Defensive Third - Patient possessions seem a bit counterintuitive at first glance, but they reflect that sustained, structured build-ups from deep tend to methodically work into dangerous positions, even if they do take a little longer
- Middle Third – Direct possessions (~20% shot rate) reflect quick transitions and counter attacks in the middle of the field
- Middle Third – Short possessions (6.3% shot rate) reflect midfield exchanges that don’t lead anywhere
- The xG per possession ranges from 0.006 to 0.023 across the possession types. Some of the highest quality chances come from possessions the get to the attacking third of the pitch quickly

Lineup Composition & Formation Analysis

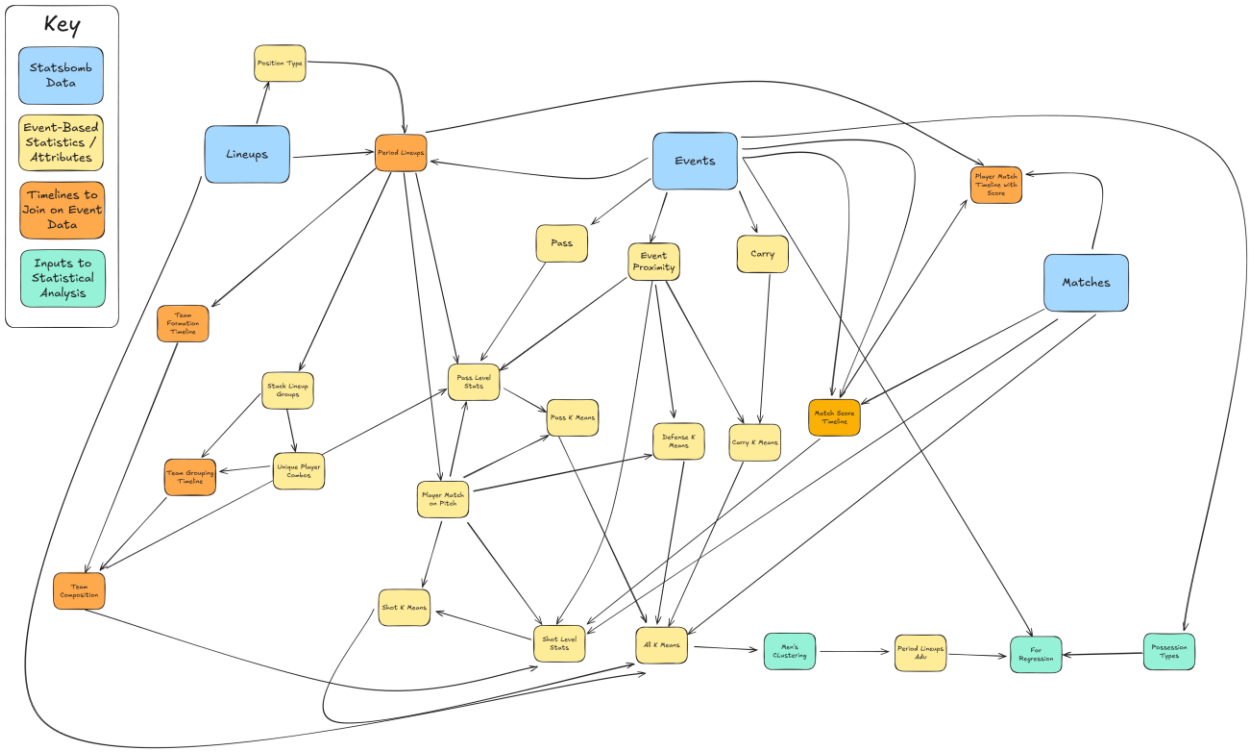
The lineups.parquet file had inconsistencies, such as end times being equal to start and end times, and was difficult to join on, with the start and end period being in the different columns. We decided to create a revamped lineups file at a period level, rather than game level. We took the Starting XI provided in the lineups file, and then brought in every substitution, dismissal due to red card / second yellow, strategic shift, and period start and end from the events data. At every timestamp for these events, we re-evaluated each player in the game and their position, creating a timeline to join back on the event data. Using the positions provided in the Statsbomb lineups file, we generalized the positions into four categories: backs (defense), midfielders, forwards and goalkeepers.

Once we had these set, we could join on the event data and determine who was on the pitch at any given time. Then, we were able to identify distinct groupings of players, whether it be the entire squad, at a position level or tracking players of the same nationality. This was useful in allowing us to track this unit's performance across many games and how they matched up with the opponent's groupings of players as well. Not only was it available at a player-specific level, but we generalized the groupings based on how many types of each position were on the field and how many of each cluster for each team. Then, we were able to aggregate the time on the pitch for each unit, as well as xG, shots taken, goals scored and other statistics to feed into the regression model. The counting stats were normalized by dividing the time together on the pitch. We limited the possessions to where both squads were full strength.

Other Data Processing

We split our data processing into several different parquet files to be utilized in other queries and processes downstream. We created a batch script that executes SQL queries via the DuckDB Python package. Each query generates the parquet files that compose the datasets for the k-means analysis and the regression analysis. For sake of space, we deleted the parquet files once they were no longer needed, but the SQL that backed a specific parquet file is in a Python file with the same name.

The sequence of the parquet files created and their dependencies can be seen here:

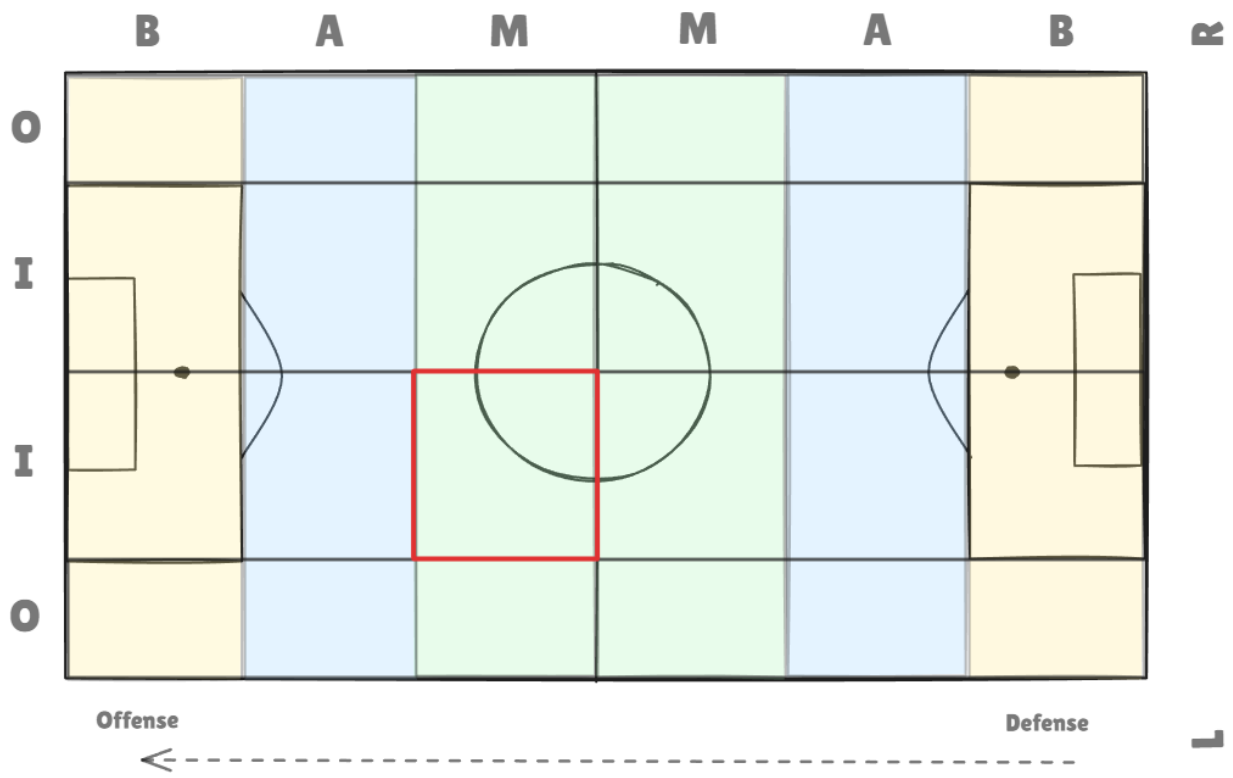


Additionally, we mapped each 2026 World Cup team to its respective group, FIFA Confederation, players that have played in each team's 2026 matches available on FBRef.com, world ranking as of January 19, 2026 (pulled from Official FIFA website), and the manager slated to lead each team this summer.

As is common with free text, it was difficult to get clean joins on player names. Some player's names were slightly misspelled or used a nickname instead. We went through the 2026 matches manually and assigned the Statsbomb ID of the appropriate players from the FBRef data. Certain countries had very limited data available in the Statsbomb dataset and were left as blank if no mappings were available.

When the Statsbomb player IDs were assigned, we narrowed in solely on possessions in the Statsbomb data where they were on the pitch. For each country, we determined a default composition of backs, midfielders and forwards based on the combination most frequently mapped for its players in the Statsbomb data. From there, we looked at the FBRef data and tagged players in each squad as starters to use as the basis for the cluster value comparison. We selected the number of players of each position based on the default formation found. Players were ranked and selected in order of minutes on the pitch in 2026 matches for their country.

We also took the dimensions and created a grid system for mapping events in certain zones on the pitch.



We divided the pitch vertically into thirds, at the edge of the penalty area and splitting the difference between midfield and the penalty area. The area closest to out of bounds was labeled B for box, the area bordering midfield M for midfield and the area between as A for arc. Horizontally, we labeled the sections O for outside the box and I for inside the box.

The remaining two indicators depend on which goal the team is attacking. For instance, in the visual above, the team is attacking to the left, so the left side of the pitch is the bottom of the pitch from the players' perspective. The left side of the pitch is labeled O for offense and right side D for defense.

The boxes were labeled in the same order: Offense/Defense + Box/Arc/Midfield + Outside/Inside + Left/Right. So, the region highlighted is "OMIL" since the team is attacking to the left. Had the team been attacking to the right, the region would have been "DMIR".

Based on the XY coordinates of the event data, we could bucket the event and its attributes to these regions. We also calculated the distance of the event from both the goal attacking and goal defending, as well as the total distance traveled by the ball between all events of the possession and the relative distance the possession gained, focusing the point of the first event to the last of the possession.

These were created in the early stages of the project and we unfortunately did not get to utilize them as much as hoped.

Player Archetype Clustering

We wanted to go beyond simple position counts when capturing the makeup of a lineup, and so we utilized a k-means clustering model that attempted to group players by their playing style. The reasoning was that a team's cluster composition (how many type A players vs type B players are on the pitch, for example) could be a better predictor of possession outcomes than simple raw position counts, because players of the same type of position, like a central midfielder, for example, can play very differently.

We characterized players using 44 features that came from the StatsBomb events dataset. It covered four domains:

- Shooting: examples include shot frequency per minute, % of shots taken in the box vs from distance
- Passing: examples include passes per minute, pass success rate, through-ball rate
- Carrying: examples include average carry distance, miscontrols, dispossession rates
- Defending: examples include duel win rate, pressures applied, blocks

We normalized all of the features using MinMax scaling. Then we clustered to prevent higher-magnitude features like raw pass counts, from overwhelming the distance calculations.

For the purposes of our analysis, we are focusing on the general clustering of the men, women, and goalkeepers. To find the appropriate number of clusters, the elbow method was used, by plotting the elbow curve and then determining where the number of clusters levels off.

For the men's league, there were four primary profiles.

- Cluster 3: Minor strength in pct_shot_first_time, percent_shot_from_q2_dist, percent_q1_pass_dist, percent_q2_pass_dist, percent_q3_pass_dist, percent_q4_pass_dist. Medium strength in pct_duel_won. High strength in pct_shots_taken_in_box, pct_of_goals_from_box.
- Cluster 2: This cluster was the least performative, showing only minor strength in percent_duals_won, pass_success_pct, and some minor strength in pass distance from the different quarters.
- Cluster 1: Minor strength in pct_shot_first_time, pct_shot_from_q3_dist, percent_q1_pass_dist, percent_q2_pass_dist, percent_q3_pass_dist, percent_q4_pass_dist. Medium strength in pct_shot_from_q4_dist, pct_shot_dual_won. High strength in pct_shot_dominant_foot, pct_shots_taken_from_in_arc.
- Cluster 0: Minor strength in pct_shot_first_time, pct_shot dominant foot, pct_shot_from_q2_dist, pct_of_goals_from_box, percent_q1_pass_dist, percent_q2_pass_dist, percent_q3_pass_dist, percent_q4_pass_dist. Medium strength in percent_shot_header, pct_shot_from_q1_dist, pct_dual_won. High strength in pct_shot_taken_from_box.

For the women's leagues there were also four clusters:

- Cluster 3: Minor strength in pct_duel_won, and percent_q1_pass_dist, percent_q2_pass_dist, percent_q3_pass_dist, percent_q4_pass_dist.
- Cluster 2: Minor strength in pct_shot_first_time, pct_shot_header, pct_shot_from_q1_dist, pct_shot_from_q2_dist, percent_q1_pass_dist, percent_q2_pass_dist, percent_q3_pass_dist, percent_q4_pass_dist. Medium strength in pct_shot_dominant_foot, pct_of_goals_from_box, pct_duel won. High strength in pct_shots_taken_in_box.
- Cluster 1: Minor strength in pct_duel_won, pass_success_pct. High strength in percent_q4_pass_dist.
- Cluster 0: Minor strength in pct_shot_first_time, pct_shot_from_q3_dist, pct_shot_from_q4_dist, percent_q1_pass_dist, percent_q2_pass_dist,

percent_q3_pass_dist, percent_q4_pass_dist. Medium strength in pct_shots_taken_from_arc, pct_duel_won. High strength in pct_shot_dominant_foot.

For the men's goalkeepers there were three clusters:

- Cluster 0: Showed the highest gk_shot_save_pct, the most strength in pct_q4_shot_xg, and some medium/low strength in pct_shot_from_gk_right, pct_shot_from_gk_left, pct_shot_from_gk_unk.
- Cluster 1: Showed strength in gk_shot_save_pct, but a notch below cluster 0, high strenght in gk_keeper_sweeper_per_minute, and medium strength in pct_shot_from_gk_right, pct_shot_from_gk_left, and gk_shot_faced_per_minute was the highest of the clusters.
- Cluster 2: Showed strength in gk_shot_save_pct, but was the lowest of the clusters, slightly lower gk_shot_faced_per_minute than cluster 1, medium strength in pct_shot_from_gk_right and high strength in pct_shot_from_gk_left.

For the women's league goalkeeper clusters, it was more difficult to select the number of clusters from the elbow chart. The most consistent notable difference between the clusters is the gk_shot_save_pct, with a notable change in cluster 1 with a high strength in gk_keeper_sweeper_per_minute, and cluster 3, which had the highest gk_shot_save_pct, also had the highest pct_q4_shot_xg, the lowest pct_shot_from_gk_right and pct_shot_from_gk_left and the highest pct_shot_from_gk_unk.

All cluster labels will be important later on when interpreting regressions results.

Regression Analyses Setup

Knowing the possession and player types, can we predict shot creation and shot quality? To answer this question, a two-stage process was developed: first, we wanted to know whether the possession produced a shot, and second, if a shot is produced, how good is it? For both stages, we built two models: one with both genders (pooled model), and one with only men's players. We will use the men's only models as the initial stages of the World Cup predictions, with the pooled models serving as checks for robustness of the models.

The regression models used four types of constructed features:

- Possession type dummies: Dummies were created for each of the eight possession types (Attacking Third – Patient technically serves as the reference category which was dropped to avoid multicollinearity). Each dummy variable captures the effect of a certain possession type compared to "Attacking Third – Patient" (the most dangerous type)

- Outfield cluster differential features (DIFF_Cx): Severe multicollinearity emerged when both offensive and defensive cluster counts were included separately, because players of the same archetype tended to appear on both teams simultaneously. For example, if one team has a lot of C3-type players, the opposing team often did as well, making OFF_C3 and DEF_C3 highly correlated with each other. So, we computed the difference: DIFF_C0 = OFF_C0 − DEF_C0, and so on for C1, C2. In theory, this should capture tactical mismatches, where the offensive team, for example, might have a numerical advantage in a given cluster archetype as compared to the defense
- Goalkeeper cluster differential features (DIFF_GK_Cx): The same differential approach was applied to goalkeeper clusters: DIFF_GK_C0 = OFF_GK_C0 − DEF_GK_C0, and DIFF_GK_C1 = OFF_GK_C1 − DEF_GK_C1. DIFF_GK_C2 is excluded as the baseline. These features capture whether the offensive team's goalkeeper archetype is more or less favorable than the defensive team's goalkeeper
- Duration: Possession duration is in minutes, reflecting how long a team maintained possession before the outcome was resolved

We removed DIFF_C3 from the models as it was the reference category. In the pooled model, IS_MENS is kept in as a binary control variable.

We computed Variance Inflation Factors (VIF) scores for all variables prior to modeling. This score measures how much the variance of a coefficient estimate is "inflated" due to correlation with other variables. As a general rule, scores above 10 indicate problematic multicollinearity. The process went as follows:

- For the full feature set, including position counts and all cluster columns, the VIFs were in the tens of thousands for position counts (they are perfectly collinear since they all sum to 11), and in the thousands for the raw cluster columns. Thus, position counts were dropped.
- Cluster columns only (OFF_C0–C5, DEF_C0–C5): Severe multicollinearity was still present in the thousands due to offensive and defensive counts of the same archetype being highly correlated.
- DIFF variables (DIFF_C0–C5, dropping C3): The VIF scores dropped significantly. DIFF_C3 still had a high value around 15 due to correlation with other DIFF features, and dropping it resulted in all VIF scores being under 5
- GK DIFF features (DIFF_GK_C0–C2): These were better than outfield clusters, with VIFs of 11–12 before dropping DIFF_GK_C2, and under 3 after dropping it
- Final feature set:

- Pooled models (both genders): duration_minutes, IS_MENS, DIFF_C0, DIFF_C1, DIFF_C2, DIFF_C4, DIFF_C5, DIFF_GK_C0, DIFF_GK_C1, and 7 possession type dummies. C3 and GK_C2 serve as implicit references; C4/C5 represent women's archetypes that are always zero for men's matches.
- Men's-only models: duration_minutes, DIFF_C0, DIFF_C1, DIFF_C2, DIFF_GK_C0, DIFF_GK_C1, and 7 possession type dummies. IS_MENS and C4/C5 removed entirely as inapplicable. C3 and GK_C2 as implicit references.

Logistic Regression: Predicting Shot Creation

The response variable produced from the lineup composition and player clustering processes used for the model is has_shot, which is a binary indicator indicating whether the possession produced at least one shot (1) or not (0). We fit the model using scikitlearn's LogisticRegression with the class weight set to "balanced" to address the 88/12 imbalance. As discussed previously, 12% of possessions produced at least one shot. The dataset was split 80/20 into training and test sets. Stratified sampling was used to preserve shot rate prevalence in both splits.

Results (Men's Only) Table:

Metric	Value	Notes
ROC-AUC (test set)	0.624	Stable across CV folds
CV ROC-AUC (5-fold)	0.626 ± 0.002	Very low variance
Pseudo R^2	0.028	Low but significant
Accuracy	54%	Balanced weighting trades precision for recall
Recall (shot class)	0.66	Model catches most shots
Precision (shot class)	0.16	Many false positives
Training observations	430,435	Men's possessions only

Several patterns emerged from the odds ratios:

- All possession types ended up reducing shot probability relative to the Attacking Third - Patient baseline. As stated previously, this reference category is the most dangerous possession type, so these results make intuitive sense. The Middle Third – Short possession type has the strongest negative effect (OR of 0.568); stagnant midfield exchanges rarely produce shots
- DIFF_C2 has the strongest negative cluster effect (OR 0.900). Having more Cluster 2 players on offense relative to defense reduces shot probability
- DIFF_GK_C0 is negative and significant (OR 0.983), meaning that having more GK_C0-type goalkeepers on the offensive team relative to the defense slightly reduces shot probability

- Duration is not statistically significant at the individual possession level (p = 0.749), meaning that possession length alone cannot reliably predict whether a shot occurs once possession type and lineup are both controlled for
- Defensive Third - Direct has a strong negative effect (OR 0.712). These are quick clearances from deep and rarely generate shots.

Linear Regression: Predicting xG Quality

The response variable for our linear regression model is the log-transformed total_xg variable, which is the sum of shot xG values from a shot-producing possession. We applied a log transformation as the raw xG values were heavily right-skewed (5.5 on full distribution). After transformation, the skew reduced to 0.14. We fit the model just on shot-producing possessions (n = 65,103 for men's). Possession without shots have, by definition, zero xG, and so zero-inflation would have dominated a model using the full dataset, not just shot-producing possessions.

Results (Men's Only) Table:

Metric	Value	Notes
R^2 (test set)	0.012	Low but statistically significant
CV R^2 (5-fold)	0.012 ± 0.003	Very stable
RMSE (log scale)	1.121	
F-statistic p-value	≈0.000	Model significant overall
Training observations	52,082	Men's shot possessions only

Several patterns emerged from the analysis:

- DIFF_C1 is the strongest positive cluster effect (exp coef 1.083), meaning that having more Cluster 1 players on offense compared to the defense tends to lead to higher xG quality.
- DIFF_GK_C1 (exp coef 1.062) and DIFF_GK_C0 (exp coef 1.058) are both positive and significant. Having a favorable goalkeeper archetype mismatch is associated with higher xG quality.
- Duration is positive and significant (exp coef 1.023) for men's matches; longer possessions are associated with modestly higher xG quality
- Middle Third - Direct has the strongest positive possession type effect (exp coef 1.029). Direct progressions through the midfield are associated with slightly higher xG per shot.
- Several possession type features and DIFF_C2 are not statistically significant at the possession level once cluster composition is controlled for

The R^2 values between 0.010 and 0.016 are low. However, we do not believe this is a model failure. The quality of shots are usually evaluated at the event-level, and often include factors like positioning, density of defenders, etc. We cannot capture these factors at the possession level; rather, what our model identifies is the share of xG variance that is attributable to systematic tactical and lineup factors. It is small, but not zero, and statistically real. In other words, this model tells us that information about the lineup and type of possession can give us modest but genuine information about shot quality. What it cannot tell you is whether a shot from a specific position is a good shot. This is an important distinction as it clarifies what kind of question a possession-level analysis can actually answer: one about patterns across many possessions, not outcomes from a single one.

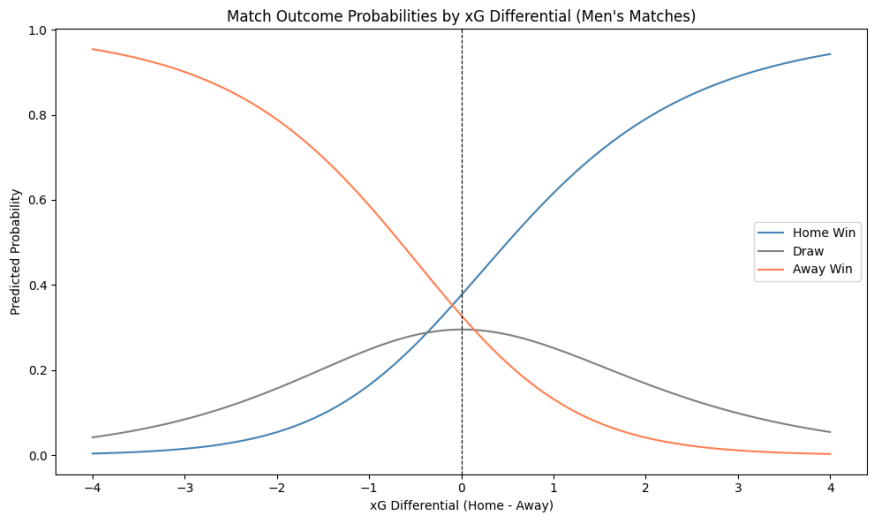
Match Outcome Model

So far, we have quantified possession-level xG. Next, we are interested in seeing whether xG differential can predict math winners. This work will serve as the bridge between our possession-level analysis and predicting upcoming World Cup match outcomes. The xG differential from a match (home team xG – away team xG) shows which team generates better chances, even if those chances weren't necessarily converted.

We aggregated xG from the possession level by summing all possessions per team per match, giving us total xG. Home and away teams were taken from StatsBomb's matches.parquet file which contained their respective ids. The xG differential showed clear, and perhaps obvious, results: home wins average +1.03 xG differential, draws average +0.15, and away wins average –0.71. We predicted the match outcomes using a multinomial logistic regression model with xG as the only feature. Away wins served as the reference. We fit the model on nearly 3,000 men's matches. The model's metrics are summed up below:

Metric	Value	Notes
CV Accuracy (5-fold)	58.2% ± 0.7%	Men's matches
Pseudo R^2	0.156	Single-feature model
Draw coefficient (vs away win)	0.742	$p \approx 0.064$
Home win coefficient (vs away win)	1.408	$p \approx 0.000$
Naive baseline (always predict home win)	≈45%	Model adds ~13pp

The model's output can be described as predicted outcome probabilities as a function of xG differential:



At a differential of 0, the model predicts roughly a 35% home win, 30% draw, and 35% away win. This is a slight home advantage. As the xG differential increases towards +4 (indicating strong home dominance), the home win probability approached 92%. The draw probability peaks around 0, and falls symmetrically toward zero at the extremes of the graph, reflecting the intuitive idea that draws happen most often when the two teams are evenly matched. Notably, there is a limitation: draws are usually unpredicted by models based on xG. For one, the recall on draws is near 0 in classifications. It's a well-documented issue: draws still happen even if one team might be dominating in xG, due to the high variance of shot conversion. For example, a team could generate 2.5 xG and actually convert none of them, while the opponent scores from their only 0.3 xG chance. So, it is a useful model for identifying wins and losses, rather than predicting draws.

World Cup Comparison Dashboard

Using our data, we created an app through Streamlit that allows the user to select two teams that will be in the 2026 World Cup and show a comparison of the team's profiles. Utilizing the players mapped to each country, we brought in each of the groupings those players have been associated with and retrieved the coordinates of every shot the grouping has taken and faced. We created heat maps that are broken out by two meter bins, displaying the shots these country's players have been associated with based on xG per minute. The pitches are flipped to align the opponents' offensive and defensive regions to identify potential advantages. Defensive metrics are blue and are negative for the sake of the visualization. Offensive metrics are in red.

Behind the scenes, our xG algorithm is calculated against the two teams that are selected to give an expected value for each team.

Conclusion / Lessons Learned

While data preparation and engineering is critical, we spent time setting up metrics we didn't end up using. Even though this time was productive and these stats could be leveraged down the line by other students, we could have taken the time to enhance the dashboard instead. I (Tyler) was guilty of some scope creep and should have re-evaluated their viability to the end goal more frequently throughout the project.

Additional Data Sources Used

- Manually scraped the following sources into CSV files:
  - 2026 FIFA World Cup Qualifying information from Wikipedia here
  - 2026 FIFA World Cup Information from FBRef.com
    - Game-level statistics from 2026 matches for each of the teams qualified to play in the 2026 FIFA World Cup, selecting each team here
  - January 19, 2026 edition of the Official FIFA World Rankings
    - Current rankings found here